The Civic Capture Audit: A Pre-Registered Protocol for Locating Deployed Systems on the Pressure–Drift–Capture Ladder

Daniel Hofheinz

daniel@danielhofheinz.com · ORCID 0009-0009-5237-5887

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Abstract

Three companion papers prove the pressure, the trap, and the loop: satisfaction-optimal deference statically degrades collective epistemic capacity on contested claims; gradient ascent on measured satisfaction makes the degradation dynamical and bistable, with capture, hysteresis, and telemetry blindness; and the closed loop the deference knob lives inside admits knob-free certificates, gain ceilings, and identification from contested-stream panels. This paper specifies the audit that locates a deployment on the resulting ladder. It codifies a five-condition decision rule whose verdicts are tier-licensed by construction; assembles nine instruments mapping conditions to estimands; imports the loop-identification estimators and the plug-in certificate margin as both a corroborating instrument and the protocol's only exculpatory finding, a clearance; and fixes per-condition thresholds, power, ethics, and kill conditions under registered-report discipline. The registered decision, coverage, causal-obligation, and clearance core is machine-checked in Lean 4. Simulated operating characteristics over forty synthetic deployments spanning all four ground-truth tiers give exact-tier accuracy 40/40 with zero verdicts above the true tier and zero false certifications; misspecification and boundary-stress batteries assert the conservative halves – no false capture classification, no false certification, abstention as the designed gray-zone behavior. Existing evidence documents mechanisms consistent with civic misalignment pressure; whether any current deployment exhibits civic drift or civic capture is an open empirical question – this protocol exists to answer it.

1 Motivation and evidence status

A research program that proves a failure mode owes its readers the test that would find it in the field. The companions establish, in order, that per-interaction satisfaction optimization degrades collective epistemic capacity on contested claims – the pressure; that the coupled operator–population system is bistable, so the degradation arrives as basin membership, with a free prevention price and strictly positive release and restoration prices – the trap; and that the feedback loop containing the deference coordinate admits identification and knob-free certification from contested-stream panels alone – the loop. What none of them does is say whether any deployed system has crossed any of these lines. This paper is the instrument that says it, or – with equal discipline and equal standing – declines to.

Role. The companions prove the pressure and the trap (Hofheinz, 2026a,b) and certify the loop (Hofheinz, 2026c); this paper specifies the audit that locates a deployment on the ladder. The boundary is strict: the dynamics paper's battery carries estimands and probes at theory grain; this protocol owns corpus construction, label acquisition, experimental arms, longitudinal estimation, intervention design, decision thresholds, and reporting standards; policy translation is deferred to a separate document.

Evidence status (pressure tier only). Three recent results corroborate the incentive premise without licensing any higher verdict. In three large-scale experiments ($N = 76{,}977$; 19 models; 707 political issues), the post-training and prompting methods that most increased AI persuasiveness systematically decreased factual accuracy (Hackenburg et al., 2025): optimizing for influence degrades the truth channel. Across eleven models and three preregistered experiments ($N = 2{,}405$), sycophantic systems affirmed users far more often than humans do, and a single sycophantic interaction increased users' conviction of being right while raising trust, perceived quality, and intention to reuse – the harm and the engagement arrive through the same channel (Cheng et al., 2026). And in two studies ($N = 246$ and 452), AI assistance improved reasoning performance while inflating self-assessment, with *higher* AI literacy associated with *lower* metacognitive accuracy (Fernandes et al., 2026): the confidence–warrant gap widens precisely among those who know the most about AI. Together with the sycophancy and reward-overoptimization cluster (Sharma et al., 2024; Gao et al., 2023; Williams et al., 2025), this documents mechanisms consistent with civic misalignment pressure. Whether any current deployment exhibits civic drift or civic capture is an open empirical question; this protocol exists to answer it.

The paper proceeds in audit order. Section 2 imports the tiers and formal objects and restates the decision rule, including the loop layer that gives the protocol its only exculpatory finding. Section 3 maps nine instruments to conditions; Sections 4–7 specify them at protocol grain – statics, dynamics, intervention, laundering – in the order a real audit would execute them. Section 8 reports simulated operating characteristics: the power analysis behind the registered thresholds, including the misspecification and boundary-stress batteries that fix the protocol's behavior when its own model is wrong. Section 9 states the access tiers, their claim ceilings, and the adoption mechanism that closes the gap between them; Section 10 fixes pre-registration, ethics, and the kill conditions under which the program's mechanism is itself disconfirmed.

2 Imported preliminaries: tiers, objects, and the decision rule

The protocol invents no vocabulary. Its tiers, formal objects, and theorems are the companions'; what it adds is the discipline that turns them into verdicts. Only the decision rule is restated in full, because everything downstream – the instruments, the thresholds, the reporting language – hangs from it.

The three tiers – *civic misalignment pressure*, *civic drift*, *civic capture* – and the formal objects $(\beta, D, I, \lambda_0, c, v, \zeta, \rho_{\mathrm{cure}}, T^\star, \mathrm{Cov}_\pi(A, B), \mathrm{Var}(D), \mathrm{AC}_1(D), \mathrm{mg}(\tau))$ are defined in the companions (Hofheinz, 2026a,b); the loop parameters and the certificate margin are defined in the loop paper (Hofheinz, 2026c). Three rules bind throughout: verdicts do not transfer across tiers; no tier may be instrumented through satisfaction telemetry alone, and capture diagnosis requires telemetry-independent stratified audits by construction; and the deployment-status sentence above is the ceiling of empirical claims.

The decision rule. The rule decomposes the capture theorem into its load-bearing premises, one condition per premise, so that a verdict is never a holistic judgment: it is a conjunction whose every term has an estimator, a threshold, and a falsification. Each condition is pre-specified with its positive criterion and threshold before data collection, and disconfirmation of any condition is a publishable finding of equal standing.

#	Condition	Formal object	Established by
C1	**Evidence dilution.** On a substantial share of the contested stream, deference reduces evidence uptake below the audit-estimated optimum ($h_\tau(\beta) > 0$), and the claim mix sits below the exposure threshold ($t < T^\star$).	Dilution bracket; $T^\star$ classification	Labeled-claim corpus; stratified belief audit
C2	**Satisfaction selection.** Measured satisfaction, trust, engagement, or preference increases in deference at the operating point.	$U' > 0$ at deployed $\hat{\beta}$	Randomized interface arms
C3	**Capacity loss.** Higher deference worsens accuracy, calibration, agency, false-balance resistance, or group epistemic equality.	$C' < 0$ components; $\mathrm{MG}(0)$; $\mathrm{mg}(\tau)$	Same audit, capacity panel
C4	**Dynamic feedback.** Deference today raises residual disagreement tomorrow and tomorrow's marginal reward to further deference; the fitted trajectory lies in the captured basin.	Ratchet–manufacture pair; map fit; position relative to Γ	Longitudinal cohort; interior-window drift-law test; map fit
C5	**Laundering.** The most-harmed groups' worsening error is unpenalized – or rewarded – by satisfaction telemetry, after survivorship adjustment; the harmed group's telemetry weight tends to zero.	Decoupling theorem signature; churn twin	Decoupling-wedge analysis with IPW correction

Verdict mapping: C1–C3 license **pressure**; C1–C4 license **drift**; C1–C5 license **capture**. No partial set licenses a higher verdict. The canonical capture statement: *a deployed system is in civic capture on a domain when, on its contested stream, deployed optimization is pulled toward higher deference; that deference operates in the evidence-dilution regime and measurably degrades collective epistemic capacity; the fitted trajectory lies in the captured basin; and satisfaction telemetry no longer registers the harm to the groups bearing most of it. Short of the full set, the conditions license only the lower verdicts: pressure for the static triad, drift with the dynamic condition added.*

Population predicates and the identification bridge. The tier definitions and the statistical decision events are distinct objects. Let E, S, K, F, B, L denote, respectively, the population propositions of evidence dilution, satisfaction selection, capacity loss, harmful feedback, captured-side basin membership, and laundering. The canonical predicates are

$$\mathsf{Pressure} := E \wedge S \wedge K, \qquad \mathsf{Drift} := \mathsf{Pressure} \wedge F, \qquad \mathsf{Capture} := \mathsf{Drift} \wedge B \wedge L.$$

Thus captured-side basin membership is a necessary component of capture, not a synonym for the full verdict. The gate logic is sound only through the explicit identification implications

$$C1 \Rightarrow E, \quad C2 \Rightarrow S, \quad C3 \Rightarrow K, \quad C4 \Rightarrow F, \quad C4 \Rightarrow B, \quad C5 \Rightarrow L.$$

Conditional on those six bridges, C1–C3 imply the population pressure predicate, C1–C4 imply drift, and C1–C5 imply capture. These definitions, the hierarchy implications, and all three conditional soundness statements are machine-checked in `PaperIV/DecisionRule.lean`; the bridges themselves remain empirical identification obligations of the registered instruments.

At registered-instrument grain the implication has three separate obligation layers. Let

$$(\theta_1, \theta_2, \theta_3, \theta_{4m}, \theta_{4r}, \theta_{5d}, \theta_{5t})$$

be the true scalar estimands for dilution, selection, capacity loss, manufacture, ratchet, decoupling, and the telemetry-threshold contrast, and let (X_E, X_C, X_B, X_R) denote the latent truths that exposure is below threshold, the covariance regime holds, the basin classification is captured-side, and the retention design is valid. First, each positive one-sided decision transfers to $\theta_j > 0$ only on its primitive coverage event $|\hat{\theta}_j - \theta_j| \le R_j$. Second, the four non-scalar recorded outputs transfer to (X_E, X_C, X_B, X_R) only on four separately reported positive-side classification events. Third, the following construct bridges must hold on the registered scope:

$$\begin{aligned} \theta_1 > 0 \ \wedge \ X_E \ \wedge \ X_C &\implies E, \\ \theta_2 > 0 &\implies S, \\ \theta_3 > 0 &\implies K, \\ \theta_{4m} > 0 \ \wedge \ \theta_{4r} > 0 &\implies F, \\ X_B &\implies B, \\ \theta_{5d} > 0 \ \wedge \ \theta_{5t} > 0 \ \wedge \ X_R &\implies L. \end{aligned}$$

Scalar coverage does not validate a non-scalar classifier, neither layer proves construct validity, and the construct bridges do not calibrate event probabilities. Their exact composition into the six abstract arrows and the three scoped tier-soundness theorems is machine-checked in `PaperIV/IdentificationBridges.lean`. Supplying all three layers from an actual deployment remains an empirical obligation; the formalization prevents any layer from disappearing inside the word "identified."

The registered coverage event for a tier is the conjunction of its primitive events, not eleven separate marginal slogans: seven scalar coverage events and four non-scalar positive-side classification events. If their marginal failure probabilities are bounded by δ_j, the probability that any primitive event fails is at most $\sum_j \delta_j$ by the finite-family Bonferroni theorem, with no independence assumption. `PaperIV/RegisteredCoverage.lean` indexes all eleven components and proves that any supplied allocation whose sum is below the announced cap transfers both to familywise failure and to a false tier verdict contained in that union. Each registered report must state the eleven-way error allocation and resulting cap (or provide a joint calibration). The protocol's registered default is the equal allocation: a marginal failure budget of 1/220 for each of the eleven components, hence a 5% familywise cap, with the transfer at exactly these constants machine-checked as `registeredFalseReturnedTierProbability_le_fivePercent`; any substitute allocation must itself be pre-registered. The pointwise two-SE rules and classifier outputs alone do not supply those probabilities.

Registered decisions, abstention, and scope. Every primitive one-sided test returns one of three decisions at its registered radius R: positive when the guarded estimate exceeds R, negative when it is below $-R$, and abstain inside the gray zone or whenever its design/specification guard fails. For a multi-part condition, any negative component disconfirms the conjunction; with no negative component, any unresolved component forces abstention; only an all-positive component set establishes the condition. C1 conjoins the dilution test, exposure classification, and covariance-regime check; C4 conjoins manufacture, ratchet, and basin decisions; C5 conjoins the decoupling wedge, harmed-group telemetry threshold, and a valid retention/IPW identification design. The rule returns the strongest fully established tier: an unresolved C5 preserves an earned drift verdict, an unresolved C4 preserves an earned pressure verdict, and any failure inside C1–C3 licenses no tier. Every record is attached to one deployment, domain, nonempty registered contested stream, and nonempty visited operating envelope.

This three-valued decision algebra, the strongest-tier function, exact no-partial-set theorems, and conditional soundness of each returned tier are machine-checked in `PaperIV/RegisteredProtocol.lean`. C4′ and clearance are typed as supplementary findings and are excluded from the tier function, so neither can substitute for C4 by construction. A positive clearance is conditionally sound on its registered margin-coverage event and the certificate model bridge for the visited envelope; the coverage event and bridge are reported obligations, not consequences of the word "clearance."

The loop layer (C4′ and clearance). The loop paper adds a two-sided instrument the ladder otherwise lacks (Hofheinz, 2026c). From contested-stream panels, the estimators $(\hat{\lambda}_0, \hat{\eta}, \hat{g}, \hat{I}_0)$ estimate the loop parameters, and the plug-in margin $\hat{m} = \hat{v}(\hat{\lambda}_0 - \hat{g}) - 2\hat{c}\hat{I}_0$ carries safe-side semantics. **C4′ (capturability corroboration):** on its registered coverage event, $\hat{g} - \hat{\lambda}_0 > 2\,\mathrm{SE}$ establishes the positive true gain-minus-grounding contrast; the additional model bridge makes that a claim that the loop can sustain capture regardless of the current basin. It strengthens a C4 finding but cannot substitute for the basin fit. **Clearance:** on its five-parameter margin-coverage event and certificate model bridge, $\hat{m} > 2\,\mathrm{SE}(\hat{m})$ certifies that no policy on the knob can produce capture on the visited envelope – the protocol's only exculpatory finding and the only one that speaks to the loop rather than the state. The coverage event, its probability calibration, and the model bridge are reported obligations, so a positive point estimate alone is not a clearance. Clearance is stated on the visited operating envelope (Section 5).

3 Instruments

Nine instruments carry the five conditions and the loop layer, and the mapping is many-to-many by design: no condition rests on a single instrument, and the heaviest verdicts require agreement across instruments with different failure modes. The inventory is assembled in the tradition of algorithmic auditing, which established both the outside-in methods available without an operator's cooperation and the ceilings on what they can conclude (Metaxa et al., 2021), and the internal accountability processes that cooperation makes possible (Raji et al., 2020). What is added here is the licensing: the verdicts are fixed by proved theorems about a specific mechanism rather than by the auditor's judgment, so each instrument is chosen to make one premise of one theorem individually falsifiable. A fitted map can be misspecified, but a certificate margin is a proved sufficient condition and a pulse response uses no fitted-map specification – its obligations are the registered causal premises of Section 6; telemetry can be blind, but an audit-retained panel keeps churned users in measurement by design. The table names each instrument, what it estimates, and the condition it serves; the four sections that follow specify them at protocol grain.

Instrument	Estimates / establishes	Condition served
Belief-seeded prompt audit	Effective deference $\hat{\beta}$ and its personalization across types	C1, C2 support
Stratified human-subject audit	η, κ, γ_τ, harm incidence, MG(0)	C1, C3
Randomized interface arms	Satisfaction-versus-capacity contrast across default, high-deference, evidence-forward, scaffold	C2, C3
Longitudinal cohort	Drift-law test of $D_{t+1} = s(\beta_t)D_t + I$; the (β_t, D_t) path	C4
Basin certificates and map fit	Capture/calibration certificate margins and their trend; side of Γ where neither fires	C4
Loop-identification panel	$\hat{\lambda}_0, \hat{\eta}, \hat{g}, \hat{I}_0$; certificate margin $\hat{m}$ with delta-method SE	C4′; clearance
Early-warning dashboard	$\mathrm{Var}(D)$, $\mathrm{AC}_1(D)$ on stationary plateaus	Drift corroboration
Intervention experiment	Hysteresis asymmetry; $\hat{\rho}_{\mathrm{cure}}$ and $\hat{\rho}_{\mathrm{restore}}$ against closed forms	C4; behavioral basin identification
Telemetry-laundering analysis	Decoupling wedge with churn adjustment	C5

4 Static estimation

The static layer runs first because it is the cheapest, the least access-demanding, and the gate to everything else. It establishes whether the contested stream is in the mechanism's regime at all (C1: the dilution bracket and the covariance test), whether the deployment's own reward points toward deference there (C2), and whether deference is purchasing capacity loss (C3). If C1 fails the audit terminates with a verdict of *none*, and that termination is a finding, not a failure – scoped by which half failed. A dilution bracket signing non-positive clears the mechanism at its root: deference there does not dilute evidence. A covariance test signing negative while the bracket stays positive places the stream outside the distributional regime only: the stratification clauses are retracted, the accuracy channel remains live, and the verdict of *none* is the rule's conservative floor, not a clearance. Four instruments compose the layer.

(a) Corpus construction. The corpus is a panel of contested-but-adjudicable claims – claims on which groups disagree but to which independent evidence can still be assigned; indicative classes: public-health, election-procedure, economic, climate, crime-statistic, legal-rights, and politically salient factual claims – with per-claim estimation of the evidence posterior m, group prior distributions b_i, the adjudicated label θ, straddle/covariance structure, responsiveness η, and effective deference. Labels by the three strategies of the companion battery, imported: delayed-resolution claims scored forecasting-style after ground truth arrives; adversarially balanced expert panels; synthetic-injection studies in which θ is known by construction (the calibration arm of the instrument suite).

(b) Belief-seeded prompt audit. For a fixed underlying question, vary the implied user prior across frames – assertion of truth, assertion of falsity, community endorsement, skepticism of mainstream evidence, suspicion of deception – and estimate effective deference as the movement of the answer toward the user's belief at held-fixed evidence, stratified by user type (high- and low-literacy, politically committed, low domain-knowledge, emotionally vulnerable, high-trust, adversarially skeptical). The audit estimates $\hat{\beta}$, its personalization across types, and the brackets and correlations feeding MG(0) and mg(τ). The capture-relevant contrast is not "the model agrees" but "the model shifts toward user priors more strongly for users least able to verify, and that shift raises measured satisfaction while worsening accuracy or calibration."

(c) Randomized arms. Default assistant; high-deference variant (warmer, more validating, more prior-accommodating); evidence-forward variant (cites evidence, corrects gently, resists false balance); deliberative-scaffold variant (elicits reasons, separates evidence from speculation, asks what would change the user's mind). Graded deference sub-arms additionally identify the operator-side constants $(\hat{v}, \hat{c})$ with standard errors – from first and second arm differences respectively – feeding the certificate margin of Section 5.

(d) Twin metric panels. A satisfaction panel (ratings, trust, return intention, engagement, willingness to reuse) and a capacity panel (belief accuracy, confidence calibration and the confidence–warrant gap, unaided reasoning, source-checking, false-balance susceptibility, polarization, between-group error gap, within-group dispersion, belief-revision willingness), aligned with the Civic Alignment Battery's dimensions. The C2–C3 result of interest: the high-deference or default arm matches or exceeds the evidence-forward arm on satisfaction while producing worse capacity outcomes, concentrated among lower-verification users.

5 Dynamic estimation

The dynamic layer converts levels into trajectories: it asks not whether the gradient points toward harm but whether the deployment is moving along it, and on which side of the basin boundary the motion lives. Its five instruments share one cohort and one discipline – every estimand is computed on data the deployment's own telemetry does not control – and its design constraints are responses to measured identification failures rather than precautions – the order of operations a registered protocol requires.

Cohort. The spine of the layer is a multi-week-to-multi-month longitudinal cohort repeatedly using the system on civic information, estimating per period the effective deference β_t, the disagreement stock D_t (aggregate belief–evidence deviation across active contested claims), the injection rate I, the grounding rate λ_0, measured satisfaction U_t, and capacity C_t.

Drift-law test (C4), with a design requirement. Test the ratchet–manufacture pair empirically: holding the claim stream constant, higher deference today predicts higher residual disagreement tomorrow (manufacture: the coefficient on $\beta_t D_t$ in the stock regression), and higher disagreement raises the measured reward to further deference (ratchet: the observed-gradient regression on D_t). **Design requirement, established in simulation:** the test is unidentified at the attractors, where deference stops varying – corner censoring – so the protocol mandates interior-transient observation windows with designed micro-perturbations of deference (the randomized arms supply them), and the stock regression is instrumented with the lagged stock: observation noise on D enters regressor and response alike, and under dither-only deference variation the interaction column $\beta_t D_t$ is nearly collinear with D_t, so errors-in-variables attenuation can flip the manufacture sign on individual cohorts – the lagged instrument removes it, the observation noise being independent across periods. Under this design, both signs are recovered in 30/30 simulated cohorts at $T = 250$ periods with 4% dither and 5% observation noise (Section 8).

Basin position (C4): a stratified instrument. The layer reports basin membership at three grades of evidence, and the grade is part of the finding. Stratifying by evidence quality is the same discipline the protocol already applies to access tiers, one level down: coverage is preserved and certification is gained, rather than traded.

Certified core. Two one-sided sufficient conditions decide membership without locating Γ at all. Capture is certified when $\hat{\beta} > \hat{\beta}^{\dagger}$ and $\hat{D} \geq \hat{D}^{*}_{g}(\hat{\beta})$, by characteristic capture, which carries no upper bound on the stock and so applies at any observed level; calibration is certified when $\hat{\beta} \leq \hat{\beta}^{\dagger} - \rho$ and $\hat{\alpha}\big(\hat{D} - \hat{D}^{*}_{g}(\hat{\beta})\big) \leq \hat{\varepsilon}^{*}$, by the sub-saddle trap at its explicit constants (Hofheinz, 2026c). Both are evaluated with delta-method standard errors on the same estimates that carry the certificate margin, and both certify only at 2 SE, so their errors are one-sided in the safe direction on their registered coverage events, exactly as for $\hat{m}$.

Model-dependent classification, with the applicability gate. Where neither certificate fires, the fitted planar map is simulated forward and the state's position and velocity relative to Γ are reported *labeled model-dependent*: the verdict is conditional on the fitted specification and inherits its misspecification risk. Simulated accuracy: 30/30 across worlds seeded on both sides of Γ. **Gate, pre-registered:** basin classification is defined only in the convergent regime, so the fitted parameters must satisfy $\hat{\alpha} < \alpha_{\text{mono}}(\hat{\theta})$ before any separatrix quantity is reported (Hofheinz, 2026b). Deployments failing the gate, or failing to reach a certified trapping neighborhood within the pre-registered horizon, are classified *unresolved/oscillatory* – a third outcome alongside the two basins, not a missing datum. The companion shows that above the threshold cooperativity alone no longer forbids sustained oscillation, and exhibits an exact period-two orbit satisfying every standing assumption; oscillation detected in a gated-in deployment therefore falsifies the mechanism, while in a gated-out deployment it is the regime the theory permits.

Fitted-classifier risk record. The residual-band classifier reports a signed basin score and three nonnegative radii for fitted-map error, visited-envelope error, and state-estimation error. Its positive-side radius is their sum; a positive classification requires the guarded estimated score to exceed that total. If the signed-score error decomposes into the three named components and their absolute errors remain inside their radii, the positive decision implies a positive true basin score exactly. If the three component failure probabilities are bounded by $(\delta_M, \delta_E, \delta_S)$, the false-positive probability is at most $\delta_M + \delta_E + \delta_S$, without independence. These deterministic and probability transfers are machine-checked in `PaperIV/BasinClassification.lean`; estimating the radii and validating their marginal bounds remain obligations of the fitted-map study.

Behavioral adjudication of the residual band. States on which neither certificate fires and the model-dependent classification sits close to Γ escalate to the intervention module (Section 6), which identifies membership by response to a transient pre-registered perturbation and uses no fitted map. Certificates decide the easy cases cheaply, the fitted map covers the middle at stated risk, and the pulse adjudicates the hard middle at higher cost and invasiveness: an escalation ladder whose rungs fail through different mechanisms, which is what the instrument design intends throughout. Both certificates are conservative by construction, so tightening their constants – the reachable-interval derivative maximum in place of the global Lipschitz bound, and the factor-of-two slacks inside ε^{*} – narrows the residual band directly, and is the cheapest available improvement to C4's coverage.

The margin is the drift-tier estimand. Drift is a claim about motion, not position, so the protocol reports the *distance to each certificate* and its trend across the panel, not only the verdict. Both distances are continuous functions of the same estimates. A capture-certificate margin that shrinks monotonically across periods is evidence, at its estimation uncertainty, that the deployment is moving toward the basin boundary – without ever locating Γ, and without the misspecification risk that attends any fitted position. This is C4's estimand of record for the drift tier, and it is the more actionable quantity in any case: a regulator reads how fast provable safety is eroding, rather than which side of an estimated curve the system currently sits on.

Loop-identification module (C4′ and clearance). The three estimators of the loop paper, at audit grade: $\hat{\lambda}_0$ as retirements over claim-period exposure; $\hat{\eta}$ from within-claim contraction ratios regressed on $(1-\beta_t)$; $(\hat{g}, \hat{I}_0)$ as the OLS slope and intercept of arrival mass on lagged stock. Across thirty seed-pinned panels under 10% multiplicative label noise, the estimators deliver mean relative errors of 2% ($\hat{\lambda}_0$), under 1% ($\hat{\eta}$), 23% of the λ_0 scale ($\hat{g}$), and 4% ($\hat{I}_0$) at $T = 600$; the corresponding ninetieth percentiles are 3%, 1%, 54%, and 10%. These finite-sample values are simulation-law frequencies, not field guarantees. The certificate margin $\hat{m} = \hat{v}(\hat{\lambda}_0 - \hat{g}) - 2\hat{c}\hat{I}_0$ carries the safe-side rule with a delta-method standard error propagating all five inputs $(\hat{v}, \hat{c}, \hat{\lambda}_0, \hat{g}, \hat{I}_0)$ – the operator-side pair estimated from the graded arms of Section 4. With $\hat{\vartheta} = (\hat{v}, \hat{c}, \hat{\lambda}_0, \hat{g}, \hat{I}_0)$ and its joint covariance matrix $\widehat{\Sigma}$, the registered variance is the full quadratic form $\nabla m(\hat{\vartheta})^\top \widehat{\Sigma} \nabla m(\hat{\vartheta})$, where $\nabla m = (\lambda_0 - g, -2I_0, v, -v, -2c)$. The implementation propagates the joint $(\hat{v}, \hat{c})$ covariance from their shared randomized-arm means and uses a joint HAC covariance for $(\hat{\lambda}_0, \hat{g}, \hat{I}_0)$, retaining both retirement–content and OLS slope–intercept terms; the two blocks are independent only because they use distinct randomized-arm and loop-panel slices. A diagonal-only sum is admissible only if the corresponding off-diagonal covariances are established as zero. The module additionally carries a specification guard: the arrival regression is refit with a quadratic stock term, and certification proceeds only when the curvature is insignificant. Certificates are envelope-scoped: the certified statement covers the operating range the panel visited, and proximity of the deployed state to the envelope's edge is itself a reported quantity. Certify only when the guard passes and $\hat{m} > 2\,\mathrm{SE}(\hat{m})$. **Identifiability requirement, measured in simulation:** certificate power requires wide deference variation and panel length – at $T = 1200$ with strong β-variation, the full-covariance five-parameter rule certified 19/20 genuinely safe loops while issuing 0/20 false certifications on capturable loops; both the v–c and g–I_0 cross terms were active in all 40/40 trials, changing the variance by as much as 91% of its diagonal contribution. The event $|\hat{m} - m| \leq 2\widehat{\mathrm{SE}}(\hat{m})$ held in 40/40 of these trials; its exact 95% binomial interval under the seed-sampled simulation law is $[91.2\%, 100\%]$. The single abstention is the designed cost of honest uncertainty; at short panels with narrow variation, $\mathrm{SE}(\hat{g})$ dominates the margin and the rule abstains. Here "safe-side" names the exact implication on the registered coverage event. The 40/40 coverage count is empirical, while the estimated-SE event probability retains the loop paper's stated asymptotic assumptions; neither is a field-population coverage guarantee.

Early-warning dashboard (Φ_2). Rolling variance and lag-1 autocorrelation of measured disagreement on the contested stream, estimated on stationary plateaus. These are critical-slowing-down indicators, imported from the early-warning literature for critical transitions (Scheffer et al., 2009; Dakos et al., 2008), where rising variance and autocorrelation precede fold bifurcations in systems whose governing equations are not known; here the governing equations are known, so the indicators are corroboration for a fold the theory already locates rather than the primary evidence for one. Simulated power under 10% observation noise: 30/30 detection at $0.9\,I_{\text{fold}}$ against a $0.4\,I_{\text{fold}}$ baseline, with 0/30 false triggers in fixed-injection controls.

6 Intervention module

The intervention module carries the program's sharpest falsifiable prediction – the prevention–cure asymmetry – and it is the protocol's only manipulation of the deployment rather than observation of it, which is why its design is the most constrained and its ethics obligations (Section 10) the most demanding. It also closes an inferential gap the protocol's observational instruments leave open: basin membership identified by response to a transient perturbation carries no fitted-map specification risk – its obligations are the registered causal premises below – so the module and the map fit fail through different mechanisms, with no independence assumed, and corroborate jointly.

Design. The module is a two-condition experiment operationalizing the hysteresis probe. Treatments are civic-weighted objectives or interface constraints: stronger evidence weighting, anti-false-balance constraints, uncertainty calibration, mandatory source contrast, "what would change your mind?" prompts, answer delayed until the user states reasoning, and reduced personalization on contested civic claims. **Prevention condition:** apply the treatment before drift. **Cure condition:** apply the same treatment after the default system has run.

Predicted asymmetry. Per the companion's hysteresis theorem (Hofheinz, 2026b): light scaffolding prevents early – the calibrated corner is stable at every civic weight $\rho \geq 0$, so the prevention price is zero – but after sufficient default running the same intervention no longer suffices: *releasing* the captured corner requires $\rho > \hat{\rho}_{\text{cure}}$ and *restoring* calibration requires $\rho > \hat{\rho}_{\text{restore}} \geq \hat{\rho}_{\text{cure}}$, both paid transiently; reversion after removal identifies the captured basin behaviorally. The protocol reports $\hat{\rho}_{\text{cure}}$ against the closed form $\rho_{\text{cure}} = 2c(1-c) - v\lambda_0/I$ evaluated at the estimated $(\hat{c}, \hat{v}, \hat{\lambda}_0, \hat{I})$. **Both prices are reported estimands.** The two coincide exactly when the companion's crossing function Ψ attains its maximum at the captured corner, for which $c^2 I < v\eta(1-\lambda_0)(1-\eta)$ is sufficient; only coincidence licenses reading Φ_4 as basin identification rather than as escape. Every input is already estimated by the loop-identification module of Section 5, so the protocol reports $\hat{\rho}_{\text{restore}}$ alongside $\hat{\rho}_{\text{cure}}$, together with $\hat{c}^2\hat{I}$ against $\hat{v}\hat{\eta}(1-\hat{\lambda}_0)(1-\hat{\eta})$, each with a delta-method standard error. A deployment in which the two prices separate is not a gap in the protocol but a reportable finding: released trajectories there may settle at an intermediate weighted attractor – civic-weighted partial capture – rather than at calibration, and Φ_4's verdict is downgraded from restoration to release accordingly.

Pulse operating characteristics. Simulated at the benchmark: a pulse at $\rho = 1.5\,\rho_{\text{cure}}$ (the benchmark has $\rho_{\text{restore}} = \rho_{\text{cure}}$, so this is super-restoration) escapes the captured basin with a minimum pulse length of $P_{\min} = 40$ periods, with escape in 20/20 noise-jittered captured states; a sub-cure pulse at $0.5\,\rho_{\text{cure}}$ never escapes (0/20) even at ten times the length; the calibrated state is unaffected by the same pulse. Basin membership is thereby identified behaviorally, under the registered obligations below – by response to a transient, pre-registered perturbation – rather than asserted from a fitted map alone, and the two instruments (map fit, pulse response) cross-validate.

Pulse identification obligations. A positive pulse classification enters C4 only on a registered event establishing intervention fidelity (delivered equals assigned pulse), consistency of the observed trajectory with the delivered intervention, temporal stability over the target period, observability of the response signature, and an exclusion/construct bridge that rules out a relevant direct pathway producing the same signature outside the captured basin. The exact composition of those premises into the latent positive basin classification is machine-checked in `PaperIV/CausalIdentification.lean`; the premises and the classifier's error probability remain deployment-specific empirical obligations.

7 Laundering module

Laundering is capture's signature and its camouflage at once: the regime in which the harm is largest is the regime in which the operator's own dashboards are most serene, because the agreement that produces the harm is the same agreement the metric rewards. The module therefore trusts the deployment's metrics with nothing – including the measurement of their own blindness – and its central statistical instrument exists because the harmed population exits the data twice, once by weight and once by churn.

Positive signatures. Low-verification users grow more wrong while rating the model highly; larger confidence–warrant gaps coincide with higher trust; harmful belief change leaves thumbs-up and return intention unmoved; model updates that improve satisfaction worsen capacity; and the harmed group's contribution to the measured loss tends to zero – by weight-vanishing in-platform and by churn out of it. The empirical statement of capture's fifth condition: the satisfaction metric becomes decorrelated from, or positively correlated with, epistemic harm among the vulnerable group, *after adjusting for attrition*.

Survivorship correction, with a design mandate. Telemetry computed on survivors inherits the blindness it is meant to detect, and so do guardrail metrics computed on the same base (Hofheinz, 2026b). The protocol's correction is inverse-probability weighting on a fitted attrition model (Robins et al., 1994) – and its validity boundary is itself a finding. In simulation – a battery measurement, not a field guarantee – under moderate harm-dependent churn the IPW-corrected harmed-group error recovers the truth (0.328 against a true 0.338, where the naive survivor estimate reads 0.221); under annihilating churn, positivity fails in the sense standard in the causal-inference literature (Hernán and Robins, 2020) and no reweighting can recover what has left the support (0.128 against 0.337). **Audit-retained enrollment is therefore mandated, not optional** – a registered protocol choice, its causal premises the C5 retention-design obligations: panelists are enrolled before exposure and retained in measurement when they churn off the platform: platform survivorship drops out of the harmed-group estimand, which then rests on post-exit measurement itself succeeding. The audit instrument is telemetry-independent by construction – the instrumentation constraint applied to the test's own instrumentation – and the deployment's own guardrail metrics are analyzed for inherited blindness as a reported quantity, not assumed sound.

Survivorship identification obligations. A positive retention/IPW component enters C5 only with consistency, conditional exchangeability (or the registered MAR analogue), positivity, and the registered nuisance-model condition; for a double-robust analysis, at least one of the outcome and censoring models must satisfy its stated correctness condition. The censoring time must be fixed, retained measurement must actually succeed after platform exit, and uncertainty from weighting and nuisance estimation must be propagated. `PaperIV/CausalIdentification.lean` types these premises and their separate identification bridge to the latent retention-design truth; it does not assert that the fitted attrition model or retained panel satisfies them.

8 Simulated operating characteristics

This section is the registered report's power analysis, and every number in it is produced by a pinned, seed-locked battery (`derisk_audit.py`, ten suites) that re-runs from seed. The table is not a summary of the battery but its contract: the registered thresholds of Section 10 were chosen against these operating characteristics, and the protocol commits to them before any field datum exists. The simulated deployments are generated by the program's own model family where an instrument is being exercised, and deliberately outside it where an instrument is being stressed − because the property a field protocol must carry under model failure is not accuracy but the direction of its errors. Accordingly the misspecification and boundary-stress suites assert only the conservative halves, and the headline below is not the diagonal but what sits above it: nothing.

Suite	Design	Measured operating characteristic
Loop identification	10% multiplicative label noise; $T \in \{300, 600\}$	Rel. errors 2% / <1% / 23% of λ_0 / 4% for $(\hat{\lambda}_0, \hat{\eta}, \hat{g}, \hat{I}_0)$ at T=600; shrink with T
Safe-side certificate	Identifiable design: T=1200, wide β-path	0/20 false certifications; 19/20 power; 40/40 empirical two-SE coverage (exact 95% binomial interval $[91.2\%, 100\%]$)
Statics (MG, regime)	K=40 claims; panels per type	Power 1.00 at N=100/type; same-side world signs Cov < 0 correctly
Arm contrast (C2)	Two arms, $\Delta\beta = 0.1$, operating stock	Power 0.82 at N=150, 1.00 at N=600; $\hat{v} = 0.108$ (truth 0.10)
Drift law (C4)	Interior windows, T=250, 4% dither, 5% obs. noise; lagged-stock IV; structural map fit	Manufacture and ratchet signs 30/30; basin classification 30/30
Early warning (Φ_2)	Plateaus at 0.4 vs 0.9 I_{fold}, 10% obs. noise	Detection 30/30; false triggers 0/30 in fixed-injection controls
Survivorship (Φ_3)	Harm-dependent churn, IPW correction	Moderate churn: recovers 0.328 of true 0.338; annihilating churn: positivity failure documented
Pulse (Φ_4)	$\rho = 1.5\,\rho_{\text{cure}}$ vs $0.5\,\rho_{\text{cure}}$	$P_{\min} = 40$; escape 20/20; sub-cure escapes 0/20 at 10× length
Misspecification	Drifting λ_0 + heteroskedastic noise; quadratic content channel; nonlinear satisfaction	False C4 0/20; quad-channel certifications 0/20, guard fired 20/20; C2 sign recovery 60/60
Boundary stress	Seeds at 0.85–1.15 Γ; near-tie C5 reads; margins $m \in \{-0.01, +0.005\}$	False-captured 0/20 sub-Γ (super-Γ detected 20/20); C5 10/10 true, 0/10 mid-transit; certifications 0/10, 0/10

Numerical Result 8.1 (End-to-end verdict mapping)**.** Over forty synthetic deployments – ten per ground-truth tier (none, pressure, drift, capture), each instrumented with audit-grade noise – the full protocol with pre-specified thresholds returns the exact true tier in 40/40 cases, with **zero verdicts above the true tier** and zero false certifications. The confusion matrix is a perfect diagonal.

Four identification requirements. Each condition below is necessary for the protocol to identify what it claims to identify; none is precautionary, and the simulated operating characteristics degrade measurably without it. (i) *Interior windows with designed dither:* the drift-law test is unidentified at the attractors, where deference stops varying; corner censoring is broken by randomized micro-perturbations during mid-transit observation windows. (ii) *Identifiable certificate panels:* the certificate's power arrives only with wide deference variation and panel length ($T \approx 1200$); short, narrow panels leave $\mathrm{SE}(\hat{g})$ dominating the margin, and the rule abstains – safe-side by construction. (iii) *Audit-retained enrollment:* under annihilating churn no reweighting recovers the harmed group; enrollment before exposure, retained through platform exit, is the protocol's design-based fix – no reweighting estimator can substitute for it once positivity fails. (iv) *Specification-guarded, envelope-scoped certificates:* an affine fit cannot exclude super-envelope nonlinearity in the content channel, so the certificate carries a curvature guard – refit with a quadratic stock term, certify only when it is insignificant – and is stated on the visited operating envelope. The misspecification and boundary-stress suites assert only the conservative halves – no false capture classification, no false certification, abstention as the designed gray-zone behavior – and the basin classifier is the structurally fitted map throughout, never the generator's own dynamics.

9 Access tiers and reporting standards

An audit protocol that ignores who can run it is a thought experiment. This section states what each access tier supports and what it cannot, and – because the gap between the tiers is an incentive problem rather than a technical one – the mechanism by which the gap closes.

Two tiers, two ceilings. The division is the auditing literature's own, inherited rather than invented: outside-in methods reach what a deployed system does to users who can be sampled or simulated, and stop where the operator's own instrumentation begins (Metaxa et al., 2021), while what lies past that line requires the instrumented party's participation (Raji et al., 2020). With black-box public access, the protocol supports pressure-tier verdicts and risk-factor findings: sycophancy, deference, false balance, confidence distortion, and short-term belief effects are externally testable. Drift and capture verdicts are claims about deployed optimization dynamics – how satisfaction signals, model updates, belief states, and disagreement stocks co-evolve – and require lab cooperation, regulator access, or large-scale independent telemetry; public prompt tests show behavior, not the full feedback loop. The honest summary the protocol may state: existing public evidence can establish pressure and a presumption of risk; adjudicating drift or capture requires the longitudinal, stratified, satisfaction-versus-capacity instruments above, with telemetry access – and the protocol exists to make that adjudication possible. Published findings phrase conclusions only at the tier the access supports; black-box results claim at most pressure.

The adoption mechanism. The access analysis names a ceiling; the certificate names the incentive that lifts it. No laboratory is naturally motivated to grant the telemetry access that drift and capture verdicts require – the instrumented party bears the risk of an adverse finding and none of the benefit. The clearance verdict inverts that calculus: a certified no-capturable-loop finding is an earnable, positive compliance asset, and the instrumentation that supports clearance – the identification panel, the graded arms, the audit-retained cohort – is the same instrumentation that supports the ladder. The protocol's adoption path runs through the certificate: access is granted to earn the clearance, and the ladder rides in on the same instruments.

Reporting standard. Every report states the tier, the conditions established and failed, the thresholds against their pre-registered values, the certificate margin with its standard error when the loop panel exists, and the governing sentence of the program: *deference must be governed by exposure regime, evidence quality, user type, and trajectory state; satisfaction alone cannot safely choose it.*

10 Pre-registration, ethics, and kill conditions

Registered-report discipline is load-bearing here, not ornamental. The format exists because fixing hypotheses and analyses before outcomes are observed is what separates a test from a description (Nosek et al., 2018), and because reviewing the design rather than the result removes the incentive to have found something (Chambers and Tzavella, 2022). A protocol whose strongest negative verdict is an accusation of structural blindness cannot leave itself room to choose thresholds after seeing data, and a protocol whose strongest positive verdict is a certificate cannot let the certified party negotiate the margin. Everything in this section is therefore fixed before enrollment, published with the design, and binding on the analysis. The registration artifact is the versioned public release of this protocol – the tagged repository release, archived with its DOI (release v1.0.0, version DOI 10.5281/zenodo.21811399) – and any enrollment cites the version that binds it.

Registered-report discipline. Per-condition hypotheses and thresholds are fixed before data: C1 by $\hat{h}_L > 2\,\mathrm{SE}$ with $\widehat{\mathrm{Cov}} \geq 0$; C2 by $\widehat{U'} > 2\,\mathrm{SE}$ at the operating point; C3 by $\widehat{\mathrm{MG}}(0) > 2\,\mathrm{SE}$ on the capacity panel; C4 by both drift-law signs plus a basin verdict at its stated evidence grade – certificate at $2\,\mathrm{SE}$ where one fires, otherwise the gated model-dependent classification cross-validated by the pulse response – with the certificate margins and their panel trend reported in either case; C4′ by $\hat{g} - \hat{\lambda}_0 > 2\,\mathrm{SE}$; clearance by $\hat{m} > 2\,\mathrm{SE}(\hat{m})$ with the five-parameter standard error and the specification guard, reported with the visited operating envelope; C5 by the post-IPW decoupling wedge together with harmed-group telemetry weight below threshold. The power analysis is Section 8; analysis code is pinned in the program's public replication repository at submission, alongside the companion batteries (`derisk_numerics.py`, thirty-one suites; `derisk_loops.py`, fifteen; `derisk_audit.py`, ten) and the Lean 4 development (`lean/`) that machine-checks every theorem of the companion papers with no axioms beyond Lean's standard three.

Kill conditions. The three falsifiers of the companion's limitations section are imported as the protocol's kill conditions: a prevalence of $\mathrm{Cov}_\pi(A, B) < 0$ on contested streams, which retracts the pressure theorem's distributional clauses specifically – the accuracy clause and the drift dynamics are covariance-free; absence of variance/autocorrelation rise preceding deference increases; and flat harm-incidence curves. Each disconfirms the component it targets, and a disconfirming audit is a publishable result of equal standing.

Human-subjects safeguards. Ethics-board review; informed consent; harm minimization in contested-claim exposure designs, including post-session debrief and correction of any induced misbelief; data minimization and de-identification for any telemetry access.

Closing

The companions prove what can happen: that satisfaction optimization carries a pressure toward deference on contested claims, that the pressure becomes a trap with a one-way price structure, and that the trap lives inside a loop with measurable gain. This protocol determines what is happening –

with pre-registered thresholds, tier-licensed verdicts, instruments that do not trust the telemetry they audit, and operating characteristics measured before the first field datum. Its verdicts are deliberately hard to earn in both directions: capture requires five conditions to hold at once under instruments designed to fail through different mechanisms, and clearance requires a certified margin under a specification guard on an explicit envelope. The program's empirical front door opens onto an honest range of outcomes – none, pressure, drift, capture, or a clearance – and the strongest finding the protocol can deliver may yet be the last one.

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